

Which Wireless Network to Use for Medical Applications?

This is one of the most often asked questions from designers today about wireless communication technologies. There are many to choose from, ranging from discrete RF, where the designers build the modulation and the entire protocol, to newer IEEE 802.15.4 transceivers and stacks that do most of the work for you, to full-blown IEEE 802.11 WiFi networks. Here we focus on IEEE 802.15.4 and IEEE 802.11

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The mass adoption of wireless communications in the consumer and information-services markets has driven down the cost of wireless-networking infrastructure. As a result, many industries such as medical have begun to integrate wireless communications into applications.

Using wireless technologies in patient care enables automated monitoring of vital signs or other parameters. This automation reduces the risk of human error in recording readings while shrinking the overhead of taking repetitive measurements such as blood pressure or temperature. A side benefit is that response times of medical staff to adverse conditions can be improved. Another plus is that patients can be monitored regardless of their location. The end result is that wireless technologies allow more cost-effective and convenient patient monitoring.

Comparison between various technologies

Fig. 1 shows how the various wireless technologies stack up against each other in terms of range and data rate. While certainly viable for medical applications, proprietary RF has not been discussed here. There are many offerings, such as INSTON from Smart-Labs, MicrelNet from Micrel Inc. and

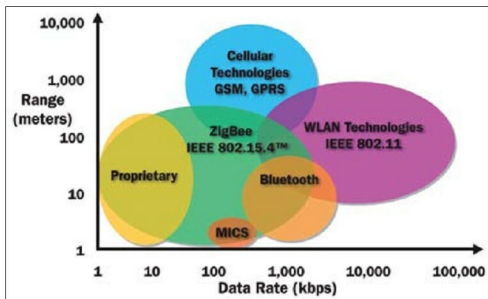


Fig. 1: Range and data rate of the various technologies employed in wireless communication applications. Proprietary solutions are those that use basic RF components to create the wireless transmitter, receiver or transceiver

Z-Wave from Zensys. Each technology offers a basic RF platform for wireless networking that is comparable to IEEE 802.15.4.

Cellular technologies offer a platform for mobile monitoring of the heart rate or ECG/EKG of a patient who is outside of a hospital or doctor's office. One drawback to cellular is the cost. The cost of the hardware is high and requires service contracts. One answer to this problem is Bluetooth. Most cellular phones are Bluetooth-enabled and allow connections to be made to headsets and other peripherals. The monitoring electronics can use a Bluetooth transceiver to connect to

the cellphone in order to upload data to a physician.

The medical implant communication services (MICS) technology is primarily intended for devices implanted into the human body to communicate with a device normally worn on the belt. It is a short-range, low-output-power technology designed to avoid interference with any other wired or wireless networking technology. It operates in the 402-405MHz frequency range, and has a maximum of 25µW output power and a data rate of up to 300 kHz.

IEEE 802.11 has been around since 1997, with the original release of 802.11